

BACLES, Romania

WMO: 154120

Lat: 44.4761N

Lon: 23.1131E

Elev: 1033

StdP: 14.16

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.9	16.4	7.2	8.3	13.7	11.1	10.0	18.9	27.9	36.9	27.4	35.7	4.5	270	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	90.1	68.6	87.5	67.8	84.7	67.3	72.2	84.0	70.4	82.3	69.0	80.8	5.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	107.6	77.7	66.5	101.4	76.3	65.0	96.0	74.8	36.5	83.8	34.9	82.3	33.7	80.8	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.6	20.3	16.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	30.2	35.9	43.2	52.2	61.5	69.5	72.1	72.6	63.2	52.6	41.9	34.4
	DBStd	16.37	8.17	8.49	8.79	8.44	5.84	5.35	5.16	5.73	7.10	7.70	7.85	7.83
	HDD50	2139	615	397	241	75	1	0	0	0	63	263	484	
	HDD65	5274	1080	814	675	389	137	18	6	10	118	386	693	948
	CDD50	3063	0	4	30	141	358	584	685	700	397	144	19	1
	CDD65	725	0	0	0	5	29	153	226	245	65	2	0	0
	CDH74	6193	0	0	0	44	210	1243	2028	2148	498	22	0	0
	CDH80	2028	0	0	0	4	23	359	704	814	124	0	0	0
	WSAvg	5.7	5.6	6.2	6.2	7.0	5.7	5.6	5.4	5.2	5.9	5.7	4.6	5.9
	PrecAvg	22.3	1.2	1.1	1.4	2.0	2.4	2.3	2.4	1.4	2.2	1.6	1.6	2.0
(15) Precipitation	PrecMax	40.1	3.1	2.8	3.4	4.9	5.4	5.5	5.2	4.2	7.0	3.6	4.2	5.9
	PrecMin	12.4	0.2	0.0	0.1	0.4	0.6	0.4	0.4	0.1	0.1	0.1	0.0	0.0
	PrecStd	5.9	0.8	0.8	0.9	0.9	1.3	1.2	1.3	1.0	1.7	1.1	1.1	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	61.6	71.3	78.7	83.1	91.2	93.0	94.5	87.9	77.4	65.8	56.7
		MCWB	46.5	50.0	54.4	60.5	67.3	68.8	71.4	68.4	65.3	60.0	54.6	47.8
	2%	DB	47.9	55.4	65.3	74.0	79.1	87.4	89.7	90.4	83.4	72.9	59.7	51.2
		MCWB	42.2	46.9	52.0	59.3	64.0	68.5	69.6	67.7	63.6	58.0	51.5	44.7
	5%	DB	44.0	50.9	60.0	69.6	76.4	84.0	86.8	87.5	79.3	68.4	55.4	47.5
		MCWB	39.5	44.7	49.2	55.7	62.6	68.3	68.6	67.1	62.6	56.2	50.5	42.9
	10%	DB	40.8	46.9	55.8	65.5	73.4	81.1	84.0	84.5	75.5	64.4	52.1	44.0
		MCWB	37.8	41.8	46.5	53.7	61.2	66.8	67.8	66.6	61.7	54.3	48.5	39.9
	0.4%	WB	46.6	51.4	57.1	62.0	68.9	72.8	76.2	74.9	68.9	61.1	55.7	50.0
		MCDB	53.0	59.8	68.5	76.5	80.4	85.2	87.6	88.0	80.0	71.2	62.3	52.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.8	48.0	53.2	59.8	65.7	71.1	72.7	71.7	66.2	59.0	53.3	45.9
		MCDB	47.3	54.5	62.5	73.4	76.1	82.9	83.8	84.0	77.4	69.3	58.0	50.7
	5%	WB	40.0	45.0	50.2	57.2	63.8	69.5	70.4	69.7	64.1	57.3	51.1	43.2
		MCDB	42.9	50.0	58.9	68.2	73.7	81.4	82.0	82.0	75.3	67.3	54.7	47.7
	10%	WB	37.7	42.6	47.7	54.8	61.9	67.8	68.0	68.0	62.6	55.2	49.2	40.5
		MCDB	40.4	46.2	54.7	63.6	71.3	79.2	80.8	79.5	73.7	62.8	52.4	43.6
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	12.8	15.3	17.4	17.6	19.3	20.2	20.0	18.1	15.5	11.0	10.6
		MCDBR	15.4	18.6	22.5	22.2	21.4	22.4	23.0	23.3	23.0	21.4	16.8	14.9
		MCWBR	10.6	12.0	12.3	11.5	9.8	9.2	8.7	8.4	8.6	10.1	10.2	10.3
	5% WB	MCDBR	12.9	17.7	21.4	21.4	20.2	21.2	21.7	20.9	20.3	20.2	13.9	15.2
		MCWBR	9.5	11.8	12.1	11.2	10.1	9.4	9.3	8.7	8.8	9.8	8.5	10.6
(40) Clear-Sky Solar Irradiance	taub		0.329	0.358	0.392	0.449	0.430	0.426	0.440	0.457	0.423	0.385	0.358	0.331
	taud		2.315	2.239	2.217	2.070	2.191	2.239	2.206	2.152	2.216	2.315	2.337	2.339
	Ebn at Noon		245	258	265	258	267	268	263	253	252	246	232	231
	Edh at Noon		26	34	39	49	45	43	44	45	39	31	26	23
(44) All-Sky Solar Radiation	RadAvg		441	697	1090	1484	1824	2041	2033	1794	1283	823	463	365
	RadStd		66	106	137	155	142	162	132	152	111	114	68	54

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (62 neighbors)		+1.10	N/A	N/A	N/A	N/A	N/A	N/A	-286	+184	N/A	

Nomenclature: See separate page